Project 1. Bank Marketing Analysis Report

Course. DSC8307 Data Mining, Modeling and Analytics

Task. Develop and evaluate a model that predicts term-deposit subscription to support the selection of customers for a marketing campaign.

Section 1. Data Exploration, Preparation and Baseline

1.1 The data

The dataset (bank-additional-full.csv) holds 41,188 telephone marketing calls. The inputs fall into four groups.

- **Client data.** age, job, marital status, education, credit default, housing loan, personal loan.
- **Current campaign contact.** contact type, month, day of week, call duration, number of calls.
- **Previous campaigns.** pdays, previous, poutcome.
- **Economic context.** employment variation rate, consumer price index, consumer confidence index, 3-month Euribor rate, number of employees.

The target is imbalanced. 36,548 calls (88.7%) did not lead to a subscription and 4,640 (11.3%) did. Accuracy is therefore a misleading measure, so this report uses ROC-AUC, PR-AUC (average precision), precision and recall

1.2 Data quality findings

- **Duplicates.** 12 rows are exact duplicates.
- **Missing values.** Six categorical columns use the label "unknown". The largest is default (8,597 rows, 20.9%), followed by education (1,731), housing and loan (990 each), job (330) and marital (80).
- **Placeholder value.** pdays = 999 means the customer was never contacted before. It covers 96.3% of rows. It is a code, not a number of days.
- **Near-empty category.** default = yes appears only 3 times.
- **Correlated indicators.** euribor3m, emp.var.rate and nr.employed correlate at 0.91 to 0.97.

1.3 What relates to subscription

Figure 1 shows the subscription rate for three key variables. Customers whose previous campaign succeeded subscribe at 65%, against 9% for customers never contacted before. Cellular contacts convert at 15% and landline contacts at 5%. March, September, October and December exceed 40%, but these months have few

calls (182 to 718). May has the most calls and the lowest rate (6%). The subscription rate also falls as the number of calls in the current campaign rises.

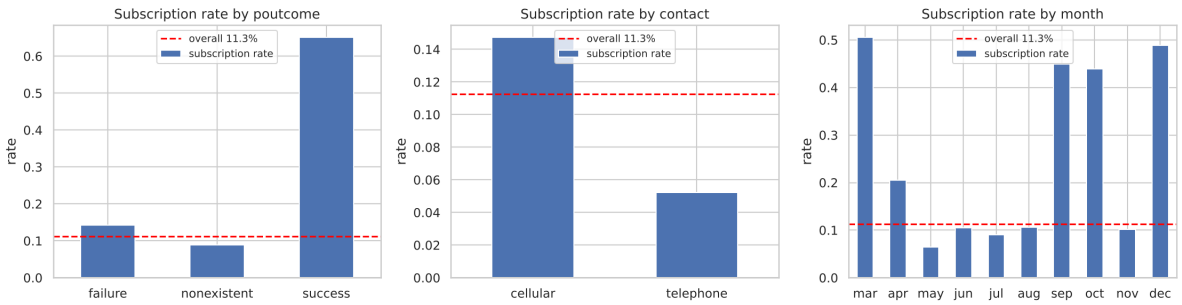


Figure 1. Subscription rate by previous outcome, contact type and month (red line is the overall 11.3%).

1.4 Data preparation and justification

Step	Justification
Remove 12 duplicate rows	They add no information and could appear in both training and test sets.
Drop duration	Call length is only known after the call ends. The dataset README says it should be dropped for a realistic model. Keeping it would be target leakage.
Replace pdays with prev_contacted (1 if pdays is not 999)	999 is a code. A yes/no flag keeps the information without distorting scaling.
Keep "unknown" as its own category	The README allows this. Not disclosing information may itself be a signal, and dropping rows would lose about a quarter of the data.
One-hot encode categoricals, standardise numericals for Logistic Regression	Logistic Regression needs comparable scales. Tree models do not. All steps are fitted inside a pipeline so cross-validation folds do not leak.
Stratified 80/20 split (32,940 train, 8,236 test)	Keeps the 11.3% positive rate in both sets. The test set is used only once, for final evaluation.

1.5 Majority-class baseline

A model that always predicts "no" was scored on the test set. It reaches 88.7% accuracy but finds no subscribers at all (recall 0). Its ROC-AUC is 0.500 and its PR-AUC is 0.113, equal to the share of subscribers. Any useful model must beat these ranking scores. Its accuracy may well be lower than 88.7%, because finding subscribers means accepting some wrong calls.

Section 2. Selection and Ranking of Approaches

2.1 Approaches selected

- **Logistic Regression.** A simple linear model that is easy to explain to the marketing team.
- **Random Forest.** Many decision trees averaged together. It can capture non-linear patterns, such as month combined with contact type.
- **Gradient Boosting.** Trees built one after another, each fixing the errors of the last. It is often the strongest method on tabular data.

Each model was scored with 5-fold cross-validation on the training data and then once on the test data. PR-AUC is the main metric because it focuses on the rare subscribers.

Model	CV ROC-AUC	CV PR-AUC	Test ROC-AUC	Test PR-AUC
Gradient Boosting	0.800	0.465	0.812	0.491
Random Forest	0.795	0.467	0.805	0.484
Logistic Regression	0.790	0.451	0.801	0.453
Majority baseline	0.500	0.113	0.500	0.113

Table 1. Model results on cross-validation and the test set.

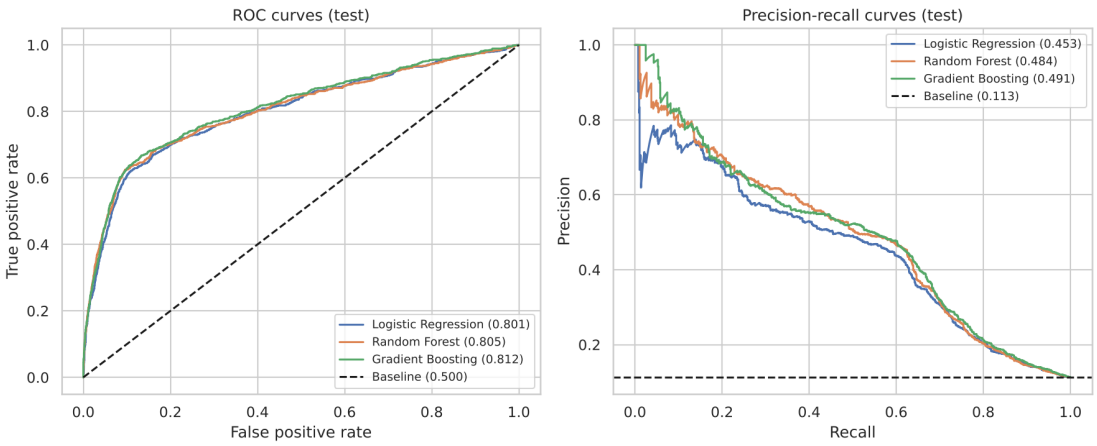


Figure 2. ROC and precision-recall curves on the test set.

All three models beat the baseline clearly. Their PR-AUC is about four times higher. The three models are close to each other.

2.2 Check on later data

The calls are in date order and economic conditions change over time. A random split mixes all periods, which may make results look better than they would be on a future campaign. As a simple check, each model was trained on the first 80% of calls and tested on the last 20%. Logistic Regression scored a ROC-AUC of 0.748 and Random Forest 0.739. Gradient Boosting fell to 0.632.

2.3 Ranking

Rank	Model	Reason
1	Random Forest	Almost as good as the best model on the random split and holds up on later data.
2	Logistic Regression	Slightly weaker on the random split but the most stable on later data. Easiest to explain.
3	Gradient Boosting	Best on the random split but drops sharply on later data.

The bank will use the model on future campaigns, so stability over time decides the ranking.

Section 3. Decision Threshold and Trade-off

3.1 The trade-off

The model gives each customer a probability of subscribing. The bank calls everyone above a chosen threshold.

- **Lower threshold.** More subscribers are found, but there are more unnecessary calls. This means more staff time and more annoyed customers.
- **Higher threshold.** Fewer wasted calls, but more subscribers are missed. This means lost deposits.

The threshold was chosen using cross-validated predictions on the training data, so the test set was not used to choose it. A simple example cost was added. Each call costs 5 units and each subscriber brings 50 units. These numbers are assumptions.

Threshold	% called	Precision	Recall	Unnecessary calls	Missed subscribers	Profit
0.05	56.8%	0.17	0.85	15,565	572	63,430
0.10	24.7%	0.31	0.67	5,631	1,217	84,075
0.15	17.2%	0.40	0.61	3,396	1,449	84,810
0.20	15.0%	0.44	0.58	2,787	1,546	83,490
0.30	11.4%	0.49	0.49	1,934	1,878	72,815
0.50	4.3%	0.66	0.25	488	2,771	39,860

Table 2. Random Forest on the training data (32,940 customers, 3,711 subscribers).

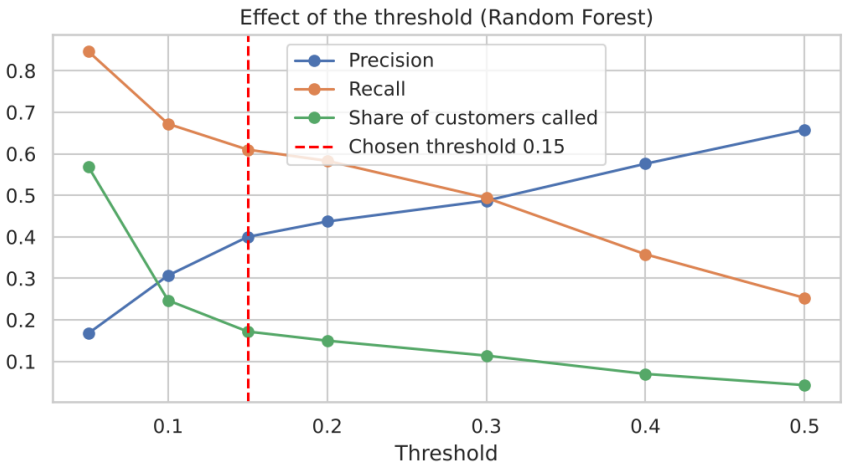


Figure 3. Precision, recall and share of customers called at each threshold.

3.2 Recommendation

The recommended threshold is **0.15**. On the test set the bank calls 17% of customers and reaches 64% of subscribers. About 42% of calls succeed, compared with 11% when calling at random.

The default threshold of 0.50 calls only 4% of customers. Its calls succeed more often (66%), but it misses three quarters of subscribers.

If a new subscriber is worth more than assumed, the bank should lower the threshold and call more people. If calls cost more, or staff time is limited, it should raise the threshold.

Limitations

- The cost figures are assumptions. The bank should redo the threshold table with its real costs.
- Results on later data are weaker, so the model should be retrained and monitored over time.
- The data only covers customers the bank chose to call.
- The patterns found are predictive, not causal.